

Autonomous Theory Construction Laboratory

Lecture 1: Iterative Research Systems

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Overview

An iterative research system is decomposed into proposal, evaluation, update, data, and attribution states.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$S_r = (T_r, \theta_r, R_r, G_r, D_r, A_r)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$c_{r+1} = P(S_r)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$S_{r+1} = U(S_r, E(c_{r+1}))$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun manual, ch. 1
- RLP-12 §§1–3
- Sandbox onboarding